
GEOSPATIAL ANALYSIS OF COVID-19 VACCINATION, A NEIGHBOURHOOD LEVEL ANALYSIS IN IRELAND

A PREPRINT

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Abstract

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1 Introduction and Background

COVID-19 vaccination has been one of the most impactful public health programs that helped mitigate the global burden of the pandemic. According to estimates published in Lancet Infectious Diseases, vaccines prevented approximately 20 million deaths in their first year of rollout, highlighting their role in reducing transmission, hospitalisation, and mortality in high-coverage settings (Watson et al. 2022). However, significant disparities in vaccine uptake have persisted across regions and population groups, reflecting deeper socio-political and structural divides.

The global COVID-19 vaccination campaign, while achieving remarkable population coverage, has revealed significant geographic disparities in uptake rates that cannot be explained by individual-level factors alone. This spatial variation in vaccination behaviour has become an important area of research, with implications for understanding health inequities and informing future public health interventions.

Spatial Patterns: Evidence from England (Dropkin 2022): Research examining spatial patterns in COVID-19 vaccination uptake has provided valuable insights into geographic disparities. A notable study of COVID-19 booster uptake across England found substantial variation, with Third Injection uptake rates ranging from 27.2% in Newham to 67.4% in Gloucestershire—representing a 40 percentage point difference between areas. This research examined area-level predictors beyond typical individual characteristics such as age, gender, ethnicity, and deprivation levels. The findings revealed that while individual socio-demographic factors remained significant predictors of vaccination uptake, considerable unexplained area-level variation persisted, suggesting that geographic factors play an important role in vaccination decision-making processes.

Research employing geographically weighted regression (MGWR) models (Chen et al. 2023) investigated how relationships between socio-demographic variables and vaccination rates vary across different geographic locations. The study found that socio-economic-demographic variables explained 83.2% of the variance in vaccination uptake patterns, with the strongest positive associations observed for proportion of population over 40 years, car ownership rates, and average household income. Importantly, this research demonstrated that the strength of associations between predictors and vaccination uptake varied significantly across geographic areas, indicating that traditional global statistical models may not adequately capture local variations in these relationships.

Barriers to Vaccination Access: Literature has identified that structural barriers, particularly geographic access, represent significant factors influencing vaccine uptake. Spatial and logistical constraints including dis-

tance to vaccination centres, limited public transport options, and regional disparities in health infrastructure have been identified as major obstacles to vaccination, even in high-income countries. Cascini et al. (2023)[5] conducted a systematic review which found that access-related challenges were particularly pronounced in rural and socioeconomically deprived areas. Similarly, Molteni et al. (2024)[6] demonstrated that physical proximity to vaccine centres remained an independent predictor of uptake in European urban and peri-urban zones.

International research has consistently documented spatial disparities in vaccination uptake across different countries and contexts. For example, de Figueiredo et al. (2021) used multilevel regression and post-stratification techniques to forecast vaccine hesitancy across the UK, successfully identifying clusters of low vaccination intent that correlated with ethnicity, economic insecurity, and urban location. In the French context, Lourenço et al. (2022)[8] found that vaccination uptake was significantly lower in regions characterized by higher poverty levels and greater migrant population density. These findings have led researchers to advocate for more granular, sub-county and neighbourhood-level modeling approaches to support more equitable public health responses. Nash et al. (2025)[9] specifically recommended incorporating travel-time variables and census-linked deprivation scores to identify vaccination “cold spots” that require targeted interventions.

Research Gap - There appears to be a significant research gap regarding spatial accessibility to vaccination sites and geographic disparities in vaccination uptake within the Irish context. This gap is particularly noteworthy given Ireland’s unique geographic and demographic characteristics. The country’s settlement pattern, which combines major urban centres (particularly the Greater Dublin Area) with rural and remote communities, creates distinct spatial challenges that may differ considerably from those documented in England or other international contexts. National statistics indicate that Ireland achieved over 90% uptake of at least one COVID-19 vaccine dose by the end of 2021, according to reporting by the Health Protection Surveillance Centre (HPSC) (“2021 - Health Protection Surveillance Centre” n.d.). However, when data was examined at the more granular LEA level, spatial inconsistencies in coverage became apparent, particularly in areas characterized by higher deprivation levels and weaker healthcare access. While researchers have applied spatial analytical approaches to study COVID-19 infection patterns in Ireland, including spatio-temporal cluster analysis, the specific application of spatial modeling techniques to vaccination services remains largely unexplored.

This background establishes that spatial variation in vaccination uptake is a well-documented phenomenon internationally, with clear evidence of structural access barriers affecting uptake patterns. However, the Irish context remains underexplored, presenting an opportunity to contribute to the understanding of geographic factors influencing vaccination accessibility and uptake in a unique geographic and demographic setting. Our work responds to those calls by conducting a spatial and demographic analysis of COVID-19 vaccine uptake in Ireland, using LEAs as the unit of observation. By merging CSO-derived demographic indicators, geospatial accessibility scores, and HPSC vaccine coverage data, the study aims to identify patterns of inequity, assess their correlates, and support data-informed, geographically targeted public health strategies.

2 Methods

2.1 Study Design

This study employs a quantitative, observational, and ecological research design to explore variation in COVID-19 vaccination uptake across Ireland. The ecological framework is suited to the research aims, as the analysis is conducted at the population level rather than the individual level, with the LEA serving as the unit of observation. This design enables the investigation of associations between area-level characteristics—such as age, education attainment, gender and deprivation scores—and aggregated vaccination rates as outcomes.

Observational designs are appropriate when experimental manipulation is neither ethical nor feasible, as is the case with vaccine uptake during a pandemic. The study does not seek to infer causality, but rather to identify statistical associations and spatial patterns that may inform future public health strategies. The ecological perspective, while limited in its ability to capture individual-level behavioural determinants, provides valuable insights into structural and contextual influences that may affect access, acceptance, or hesitancy at the community scale.

Additionally, the study incorporates a temporal dimension by analysing vaccination uptake over successive months, enabling the identification of evolving patterns and time-based disparities in vaccine coverage. This temporally extended approach complements the spatial analysis and enhances the explanatory power of the

models. The combination of spatial and temporal design elements positions this research within the growing body of data-driven public health studies aimed at understanding complex, dynamic health phenomena within real-world settings.

2.1.1 Data Sources

To address the research objectives, this study integrates multiple publicly available datasets capturing vaccination services, uptake rates, and geographic boundaries. The vaccination services data encompass three distinct service delivery models that formed Ireland’s COVID-19 vaccination network during the rollout.

Initial Vaccination Centers for COVID-19: 35 large-scale vaccination centers established specifically for the national COVID-19 immunization program were identified and catalogued. These centers represented the primary delivery mechanism during the initial vaccine rollout phases, particularly for priority populations and mass vaccination campaigns.

General Practitioners (GPs): 1557 general practice clinics (GPs) participating in the COVID-19 vaccination program were systematically identified. GP-administered vaccines represented a crucial component of community-based delivery, particularly for reaching populations with established healthcare relationships and those requiring more personalized care approaches.

Pharmacies: 1041 community pharmacies authorized to administer COVID-19 vaccines were documented as part of the expanded vaccination network. Pharmacy-based vaccination services enhanced accessibility by leveraging existing community health infrastructure and extended service hours.

2.1.2 Data Collection Methods

All vaccination center data were obtained through systematic web scraping techniques implemented on the HSE website using *rvest* in R and Selenium in Python.

Static Web Pages: The *rvest* package in R was used for scraping data from static HTML pages. For static web pages, the information is directly available in the HTML source when the page loads, making *rvest* ideal for quick and simple extraction of data from the page’s structure.

Dynamic Web Pages: Selenium in Python was utilized to handle JavaScript-rendered content, enabling access to dynamically loaded data not accessible via static scraping. Some websites use dynamic content, where information is loaded separately through JavaScript after the initial page is displayed. Selenium can simulate a real web browser, allowing the script to wait for content to load, click buttons, or scroll down the page to access hidden or delayed information. While *rvest* is faster and easier for static content, Selenium is necessary for interacting with and extracting data from more complex, dynamic sites.

All identified vaccination centers were geocoded to obtain their geographic coordinates (longitude and latitude). The Google Maps API was employed to ensure accurate and standardized geolocation across the dataset. A custom function was written to send HTTP requests to the Google API with each address and extract the corresponding latitude and longitude from the returned JSON data. This approach allowed precise control over the geocoding process and can be combined with rate limiting to comply with API usage policies.

Vaccination uptake data which is the LEA vaccination rates were sourced from table CDC47 published on the Central Statistics Office (CSO) website. This dataset provides vaccination coverage statistics at the Local Electoral Area level, representing the most granular administrative geography for which reliable vaccination data were consistently available throughout the vaccination rollout period.

LEA boundary files were obtained from Ordnance Survey Ireland (OSI), providing the official digital geographic boundaries necessary for spatial analysis and mapping applications. These boundary files serve as the foundational geographic framework for linking vaccination rates with corresponding service locations.

In parallel, socio-demographic variables—such as education, age, gender proportions and deprivation indices by LEA—were extracted from the CSO’s 2016 Census and supplementary 2022 updates where applicable.

2.1.3 Engineering Accessibility Data

Traditional accessibility measures include naïve distance-based metrics and provider-to-population ratios, Luo and Qi (Luo and Qi 2009) established the modern spatial accessibility analysis method of two-step floating catchment area (2SFCA). They demonstrated that the 2SFCA method addresses critical limitations of traditional distance-based measures by incorporating both healthcare service supply (provider capacity) and

it's demand (population need) factors simultaneously. This approach provides more accurate representations of healthcare accessibility patterns by accounting for competition among populations for limited healthcare resources. Their method first involves identifying all population locations within a reasonable travel distance of each healthcare provider, then calculation of provider-to-population ratios. In the second step, it identifies all providers within reasonable distance of each population location and sums their accessibility ratios. This dual consideration of supply and demand makes the 2SFCA method relevant for our study where both vaccination site capacity and population demand vary spatially.

Luo and Wang (Luo and Qi 2009) provided a comprehensive comparison of gravity models, enhanced 2SFCA methods, and other spatial accessibility approaches. Through their analysis in the Chicago region, it was demonstrated that different accessibility measures can produce substantially different results depending on geographic context, population distribution, and healthcare service characteristics. The study revealed that gravity models, which weight healthcare opportunities by distance decay functions, often provide more nuanced accessibility scores than simple measures. Wang's framework offered crucial direction for vaccination accessibility research by establishing important factors to take into account when choosing suitable spatial accessibility models depending on study goals and data availability.

Nevertheless, detailed demographic statistics at the sub-LEA level and, ideally, information regarding service capacity are necessary for the complete implementation of 2SFCA, and these were not available for this study. Furthermore, the approach is resource-intensive, requiring substantial geospatial processing and catchment dynamics assumptions. So, we decided not to use 2SFCA in our analysis as a result.

We used an intuitive spatial coverage measure based on proximity - travel isochrones were generated for each service location. Specifically, 10, 20, 30, and 60-minute drive-time isochrones were created for the initial vaccination center, while 10-minute drive-time isochrones were generated for pharmacies and GPs. Each LEA polygon was then intersected with these isochrones to determine the extent of overlap, representing the proportion of the LEA covered by the service catchment areas. The total overlapping area was measured to produce a naive accessibility score for each LEA. For the initial vaccination center, a more refined accessibility score was calculated by inversely weighting the overlapping areas by travel time, reflecting the assumption that closer proximity implies better access. This approach was based on the intuitive premise that any spatial overlap between a healthcare service's catchment and an LEA indicates at least some degree of physical access for residents.

Accessibility Score Calculation:

$$C_{ijk} = \frac{I_{ijk}}{T_k}$$

$$A_{ij} = \sum_{k=1}^K C_{ijk}$$

$$A_{vacc} = \frac{1}{10} \cdot A_{i,10} + \frac{1}{20} \cdot A_{i,20} + \frac{1}{30} \cdot A_{i,30} + \frac{1}{60} \cdot A_{i,60}$$

$$A_{pharm} = A_{i,10}$$

Where C_{ijk} is the coverage share, I_{ijk} is intersection area, T_k is isochrone area, and A_{ij} is the accessibility score

Applying 2SFCA—or an advanced variation—would yield a more comprehensive assessment of healthcare accessibility with more specific data and specialized resources, particularly in contrasting rural and urban situations where demand and service burden varied significantly and we strongly advise using it in subsequent research.

2.1.4 Data Merging

The process of merging datasets from multiple sources constituted a pivotal step in preparing the analytical dataset used in this study. Each dataset—vaccination data, demographic variables, and accessibility scores—originated from distinct sources and differed in structure, temporal resolution, and formatting. To ensure analytical coherence, extensive preprocessing was undertaken to harmonise identifiers, align time frames, and integrate relevant variables into a single, unified data table.

The vaccination dataset, which served had the primary outcome data contained multiple records per LEA, each corresponding to a specific reporting month. This long-format structure was transformed to suit the objectives of the analysis. For our study we only focussed on the vaccination data for people 12 years and

older and our primary outcome of interest was primary vaccine dose’s vaccination rate. The socio-demographic variables were available in wide format, with a single observation per LEA.

The critical challenge in data integration arose from inconsistencies in LEA naming conventions across the vaccination uptake data (CDC47) and the geographic boundary files (OSI). To address this systematic issue, LEA names from both datasets were pre-processed to ensure consistency in naming conventions. This preprocessing workflow addressed variations in spelling, capitalization, punctuation, and formatting that would otherwise prevent successful dataset alignment. Following this standardization process, the two datasets were joined using the `merge()` function in R, with the harmonized LEA names serving as the common linking key to establish precise one-to-one correspondence between vaccination statistics and geographic boundaries.

The accessibility variables and sociodemographic variables were stored in separate datasets and were appended via the same standardized LEA identifier matching protocol used for the primary data integration. This resulted in two final datasets – one which contained one row per LEA, with columns for monthly rates of vaccination uptake for primary, booster 1, booster 2, booster 3 doses and LEA’s geometric polygons. The second dataset contained all demographic variables, and the engineered accessibility score per LEA. These served as the analytical foundation for exploratory and modelling components of the study.

2.1.5 Model

We employed a hierarchical Bayesian modelling framework to examine determinants of primary COVID-19 vaccination uptake rate across LEAs. A total of six modelling approaches were implemented, each designed to address specific research questions, including bounded proportional outcomes, spatial autocorrelation, and nonlinear temporal trends. The models progressively build on one another, beginning with basic compositional demographic factors of age and gender and expanding to include compositional health and education metrics, spatial structure, deprivation, interaction effects of age and gender, and temporal dynamics.

All models were implemented using the `brms`, a Bayesian regression modelling interface to Stan, which supports complex hierarchical, spatial, and temporal modelling structures, and leverages Markov Chain Monte Carlo (MCMC) sampling for full posterior inference.

The primary outcome was the final rate of primary COVID-19 vaccination (bounded between 0 and 1), with booster dose data available for future extensions. The compositional nature of demographic covariates (e.g., age, sex, education, health) required Additive Log-Ratio (ALR) transformations to avoid issues of perfect multicollinearity due to unit-sum constraints.

All cross-sectional models used Beta regression with a logit link function, which is ideal for modelling proportional data as it ensures predictions remain in (0, 1) while also accounting for heteroskedasticity via a precision parameter. We did not go for zero-inflated beta regression, as our data did not contain exact 0s or 1s. This allowed robust estimation of proportional outcomes while capturing uncertainty in both the mean and spread of the distribution.

Model 1 focussed on basic demographic variables—namely age and gender—relate to the final uptake of the primary COVID-19 vaccine dose across LEAs. Bayesian inference was conducted via MCMC sampling with four chains of 2,000 iterations each (including 1,000 warm-up iterations), resulting in 4,000 post-warmup posterior draws and this was the functions’ default in R.

`Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + male_logratio`

The base model was extended to model 2 by adjusting for health, education, and accessibility to initial vaccination center. The accessibility measure was particularly relevant to the primary dose, which was more likely administered at centralized mass vaccination centres as opposed to decentralized locations like GPs or pharmacies, which were more important for booster doses. This model retained the beta regression framework and used the same MCMC setup as Model 1. Incorporating these additional variables allowed us to capture social determinants of health and logistical barriers to access, thereby enriching our understanding of factors influencing vaccine uptake beyond demographic composition alone.

`Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + edu_NoFormal_logratio + edu_Primary_logratio + edu_UpperSecondary_logratio + edu_Apprenticeship_logratio + edu_HonoursBachelor_logratio + health_VeryBad_logratio + health_Bad_logratio + health_Fair_logratio + health_Good_logratio + male_logratio + Wt_accessibility_Initial_Vacc`

Model 3: Morans'I was calculated to identify if spatial autocorrelation existed in the residuals. The presence of significant spatial clustering suggested that residuals were not independent across LEAs, violating a core assumption of standard regression models. To address this, we introduced a spatial modelling approach using Conditional Autoregressive (CAR) priors. This model built upon the covariates included in Model 2 and incorporated spatially structured random effects via an adjacency matrix representing geographical contiguity among LEAs. The CAR prior allows neighbouring areas to “borrow strength” from each other by sharing information, thus improving parameter estimation in areas with sparse or noisy data. Because spatial models are more complex and prone to convergence issues, we extended the MCMC sampling regime to 30,000 iterations per chain with 10,000 warm-up iterations, resulting in a total of 80,000 post-warmup draws. We also increased `adapt_delta` to 0.999 and set `max_treedepth` to 20 to stabilize sampling using the No-U-Turn Sampler (NUTS). This spatially informed model was crucial for appropriately accounting for geographic dependencies in vaccination rates that may arise from shared infrastructure, regional policies, or cultural clustering.

$$\text{Primary_Vax_Rate} \sim \text{age_12to17_logratio} + \text{age_18to54_logratio} + \text{age_55to64_logratio} + \text{age_65to70_logratio} + \text{edu_NoFormal_logratio} + \text{edu_Primary_logratio} + \text{edu_UpperSecondary_logratio} + \text{edu_Apprenticeship_logratio} + \text{edu_HonoursBachelor_logratio} + \text{health_VeryBad_logratio} + \text{health_Bad_logratio} + \text{health_Fair_logratio} + \text{health_Good_logratio} + \text{Wt_accessibility_Initial_Vacc} + \text{car}(\text{lea_mat}, \text{gr} = \text{lea_id})$$

To further capture the socioeconomic context beyond individual-level education indicators, the fourth model introduced a deprivation index, representing a composite measure of material and social disadvantage at the area level. This variable was added to the extended multivariable model (Model 2) and allowed us to quantify how deprivation might influence vaccine uptake independently of individual education. The deprivation index was not compositional in nature, so it was included in its raw form. This model retained the beta regression framework and standard MCMC settings. By disentangling the effects of both individual (e.g., education, health) and area-level (e.g., deprivation) determinants, this model aimed to improve explanatory power and helped identify if communities could potentially be vulnerable to lower vaccine uptake due to systemic disadvantage.

$$\text{Primary_Vax_Rate} \sim \text{age_12to17_logratio} + \text{age_18to54_logratio} + \text{age_55to64_logratio} + \text{age_65to70_logratio} + \text{edu_NoFormal_logratio} + \text{edu_Primary_logratio} + \text{edu_UpperSecondary_logratio} + \text{edu_Apprenticeship_logratio} + \text{edu_HonoursBachelor_logratio} + \text{health_VeryBad_logratio} + \text{health_Bad_logratio} + \text{health_Fair_logratio} + \text{health_Good_logratio} + \text{male_logratio} + \text{Wt_accessibility_Initial_Vacc} + \text{Deprivation}$$

With model 5 we departed from the additive structure of previous models and explicitly modelled the joint distribution of age and gender. This approach allowed us to identify group-specific patterns that could not be detected when age and sex were considered independently. For example, vaccine hesitancy might be disproportionately higher among younger males compared to older females. To capture such differential effects, we included ALR-transformed proportions of eight age-gender combinations (e.g., females aged 12–17, males aged 55–64, etc.), using females and males aged 71+ as the reference group. This model did not include additional covariates like education or health and focused solely on the interaction structure. Like earlier models, it employed beta regression with a standard MCMC sampling regime of 4,000 post-warmup draws. This model was aimed at offering a more granular understanding of population heterogeneity in vaccine uptake.

$$\text{Primary_Vax_Rate} \sim \text{alr_age_12to17_females} + \text{alr_age_18to54_females} + \text{alr_age_55to64_females} + \text{alr_age_65to70_females} + \text{alr_age_12to17_males} + \text{alr_age_18to54_males} + \text{alr_age_55to64_males} + \text{alr_age_65to70_males}$$

While the previous models were cross-sectional in nature, model 6 was used to introduce a longitudinal perspective to capture how vaccination rates evolved over time within each LEA. We observed that vaccination uptake followed a characteristic sigmoidal (S-shaped) growth curve—slow at first, then rapidly increasing before eventually plateauing. To model this trajectory, we specified a four-parameter logistic function within a hierarchical Bayesian framework. The function estimated key curve parameters: `base` (the lower asymptote or initial vaccination rate), `Asym` (the upper asymptote or final saturation level), `xmid` (the inflection point where uptake accelerates most rapidly), and `scal` (the spread or steepness of the curve). LEA-specific random intercepts were included for `base`, `Asym`, and `xmid`, while `scal` was assumed to be globally constant due to model identifiability constraints. Furthermore, we modeled `Asym` as a function of ALR-transformed covariates (e.g., age and gender composition) to understand how basic demographic characteristics shape the final uptake level.

Priors for the nonlinear parameters were carefully chosen based on empirical inspection of the data. Specifically, we set a normal(0.8, 0.1) prior on Asym, reflecting the assumption that final vaccine coverage would typically plateau around 80%, with moderate uncertainty. This aligns with observed saturation levels in many LEAs. For xmid, we used a normal(6, 2) prior, based on the understanding that the inflection point—the month with the fastest uptake—was likely to occur within the first half-year after vaccine rollout. This prior captures temporal heterogeneity while remaining centered around plausible empirical timing. Finally, we applied an exponential(1) prior (constrained to be positive via $lb = 0$) on scal, ensuring strictly positive steepness while favoring gradual curves unless strong data signals indicate rapid growth.

The model was fitted to 4,980 monthly observations across 166 LEAs using the No-U-Turn Sampler (NUTS) with four chains of 4,000 iterations (2,000 warm-up), yielding 8,000 post-warmup draws. This hierarchical model would allow us to disentangle local variations in the speed and extent of vaccine uptake over time, offering valuable insights into both structural and behavioral determinants of vaccination trajectories.

Primary_Vax_Rate $\sim$ base + (Asym - base)/(1 + exp((xmid - Month_num)/scal))

3 Results

3.0.1 Descriptive Statistics

3.0.2 Model

Model 1- Younger age groups demonstrated significantly lower vaccination rates compared to the reference category, with particularly strong negative effects for the 18-54 year age group and the 12-17 year age group. The 55-64 year old variable was a strong positive predictor in the model. The credible interval (CI) for 65-70 year olds included zero, suggesting uncertainty around this effect. Male proportion demonstrated a negative association with vaccination rates but the CI included zero. The marginal effects plots are used to understand the impact to the predictors better.

Basic Demographic Model Results			
Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	3.731	3.200	4.222
age_12to17_logratio	-0.607	-0.992	-0.226
age_18to54_logratio	-1.413	-1.755	-1.065
age_55to64_logratio	2.474	1.470	3.433
age_65to70_logratio	-0.421	-1.114	0.303
male_logratio	-1.782	-3.489	-0.057

Figure 1: Model 1 Summary

Each curve’s starting position represents the log-ratio of that age group compared to the elderly reference category, revealing Ireland’s uneven age distribution across Local Electoral Areas. The teal curve (ages 55-64) starts farthest right at approximately +1.8, indicating that these areas typically have about 6 times more people aged 55-64 than seniors ($e^{1.8}$ approx 6). This pattern is common in workforce-heavy LEAs where working-age populations dominate. Conversely, the pink curve (ages 12-17) starts in negative territory around -0.3, showing fewer adolescents than seniors in most areas—characteristic of aging communities or retirement-focused regions. Critically, this demographic baseline has direct implications for vaccination outcomes: regions with proportionally more adolescents (12-17) compared to the elderly (71+) tend to have lower vaccination rates. This relationship emerges because the compositional constraints mean that areas with higher youth populations necessarily have different age structures that correlate with vaccination hesitancy patterns. As you move along the x-axis (representing underlying gradients like socioeconomic status, education, or area deprivation), each curve shows how vaccination probability changes across different demographic contexts. The crossing points where curves intersect reveal critical thresholds where both demographic composition and vaccination behavior shift dramatically—identifying socioeconomic tipping

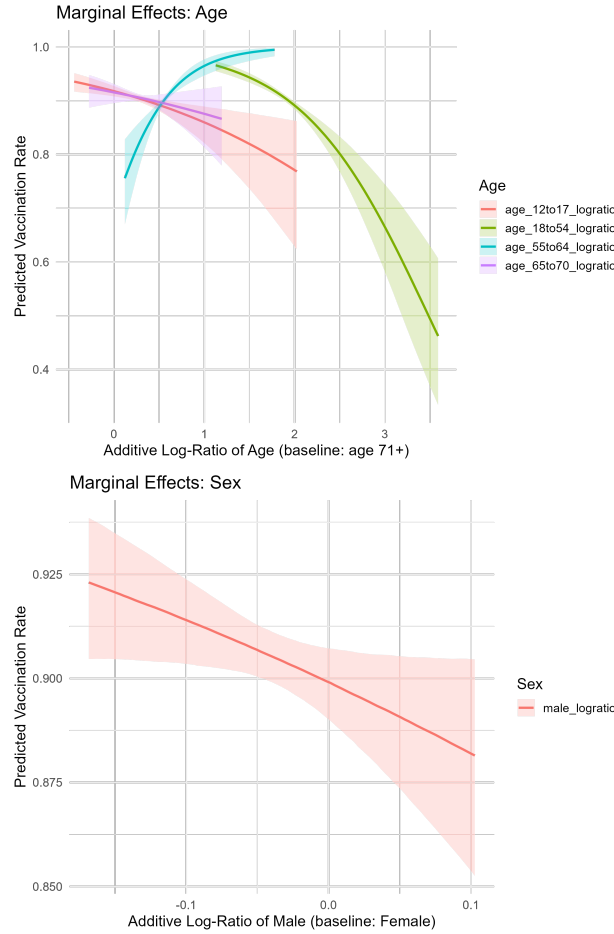


Figure 2: Base Model Marginal Effects of Adjusted Model

points crucial for targeted public health interventions. This log-ratio approach is particularly valuable because it accounts for the compositional nature of demographic data: since all age proportions must sum to 1, increasing one group necessarily affects others, and the extreme positions on the x-axis highlight areas where certain age ratios dominate (positive values) or are scarce (negative values) relative to the senior population.

Model 2 – We could see that the age effects persist but attenuated slightly while revealing additional significant associations. Upper secondary education emerged as a significant positive predictor, suggesting higher vaccination rates among populations with this educational level. Conversely, areas with higher proportions of individuals with no formal education showed lower vaccination rates. Health status variables showed no statistically significant associations with vaccination rates. The male proportion effect changed to be positive but was still non-significant. Accessibility to initial vaccination centres showed non-significant impact suggesting that geographic accessibility was not a significant barrier to vaccination uptake in our population.

Model 3 - To formally test for spatial autocorrelation, we conducted a Moran's I test on the residuals: Moran's I = 0.1906, $p < 0.001$ — indicating significant positive spatial autocorrelation in residuals (i.e., LEAs geographically close to one another tend to have similar unexplained vaccination patterns) This confirmed that standard regression models failed to fully account for spatial dependencies, prompting the use of a CAR model, introducing a structured spatial random effect term that smooths values across neighboring areas The spatial correlation parameter (car) was estimated at 0.56 (95% CI: 0.03 to 0.99), suggesting moderate positive spatial autocorrelation between neighboring LEAs, i.e., 56% of unexplained variance was attributable to spatial autocorrelation, indicating strong neighborhood-level similarities. This spatial clustering may reflect unmeasured factors such as local social networks, community attitudes, healthcare provider practices, or other geographically-clustered influences on vaccination behavior. The relatively modest spatial standard

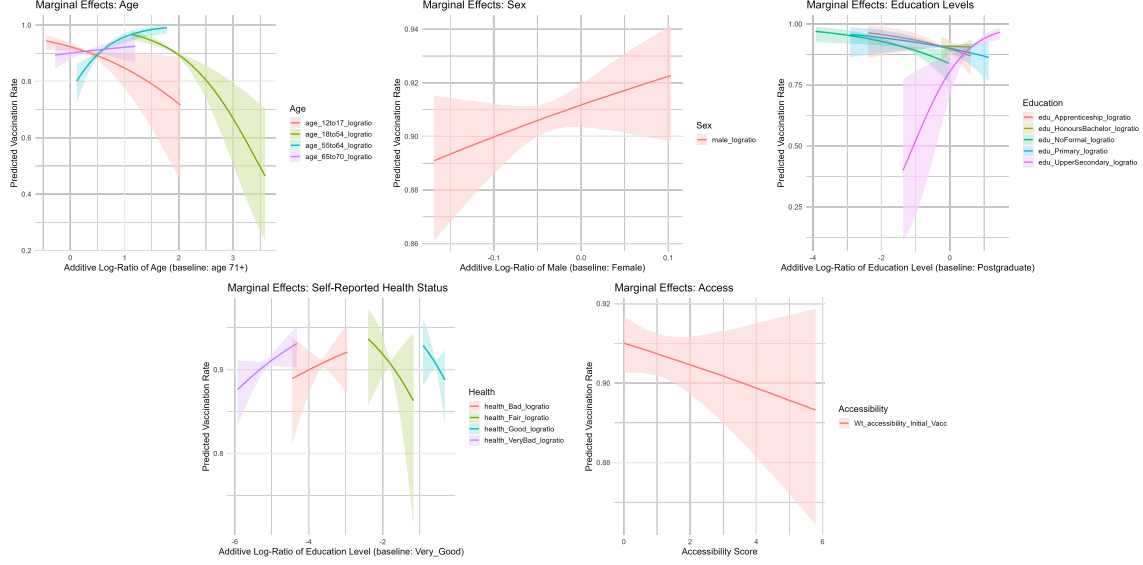


Figure 3: Marginal Effects of Adjusted Model

deviation (0.21) indicates that while spatial correlation exists, the magnitude of unexplained spatial variation is limited, suggesting that the measured covariates capture much of the systematic variation in vaccination rates. Demographic patterns remained robust: Age 55–64 still had the strongest positive effect on uptake. Upper secondary education, remains a large positive driver and healthcare proximity and healthcare outcomes continued to show near-zero effects. The consistency of these effects across spatial and non-spatial models suggests that educational disparities in vaccination uptake operate independently of geographic clustering and supply-side geographic barriers and self-reported health outcomes were not significant determinants of vaccination disparities.

The inclusion of the Pobal HP Deprivation Index, which measures relative affluence and disadvantage across Irish communities using composite scores derived from employment, education, housing quality, and demographic variables, had minimal impact on vaccination rates despite capturing broader socioeconomic conditions at the LEA level. The composite deprivation index showed a negligible association with vaccination rates, with confidence intervals including zero, suggesting that area-level socioeconomic deprivation does not provide additional explanatory power beyond the specific educational attainment variables already included in the model. This finding indicates that educational composition effectively captures the most relevant aspects of socioeconomic variation affecting vaccination behavior, and that deprivation effects operate primarily through already-measured demographic and educational pathways rather than through additional unmeasured mechanisms. The lack of significant association suggests that while the Pobal index comprehensively measures community-level disadvantage, individual educational characteristics remain the dominant socioeconomic predictor of vaccination uptake.

The age-sex interaction model revealed striking differences in age-vaccination relationships between males and females. Among females, younger age groups showed strong negative associations, with the 18–54 year group showing significant impact and among men the 65–70 year group uniquely showed a significant negative association contrasting sharply with the positive trend observed in older females, despite generally being considered a high-priority group. These results suggest that vaccine hesitancy patterns differ substantially between genders within the same age cohorts.

We found distinct patterns in COVID-19 vaccination uptake trajectories across Ireland’s LEAs using our fitted hierarchical beta regression model. The population-level upper asymptote was at 0.94 (95% CI: 0.76–1.12) on the logit scale, representing the baseline maximum vaccination rate before accounting for demographic influences. The temporal inflection point occurred at 5.61 months (95% CI: 5.57–5.66), indicating that the steepest increase in vaccination rates typically occurred during late May to early June 2021, corresponding to the period when vaccine availability expanded beyond priority groups. The scale parameter was found to be 1.36 (95% CI: 1.35–1.38), which controls the steepness of the sigmoidal curve transition and suggests a relatively gradual uptake pattern.

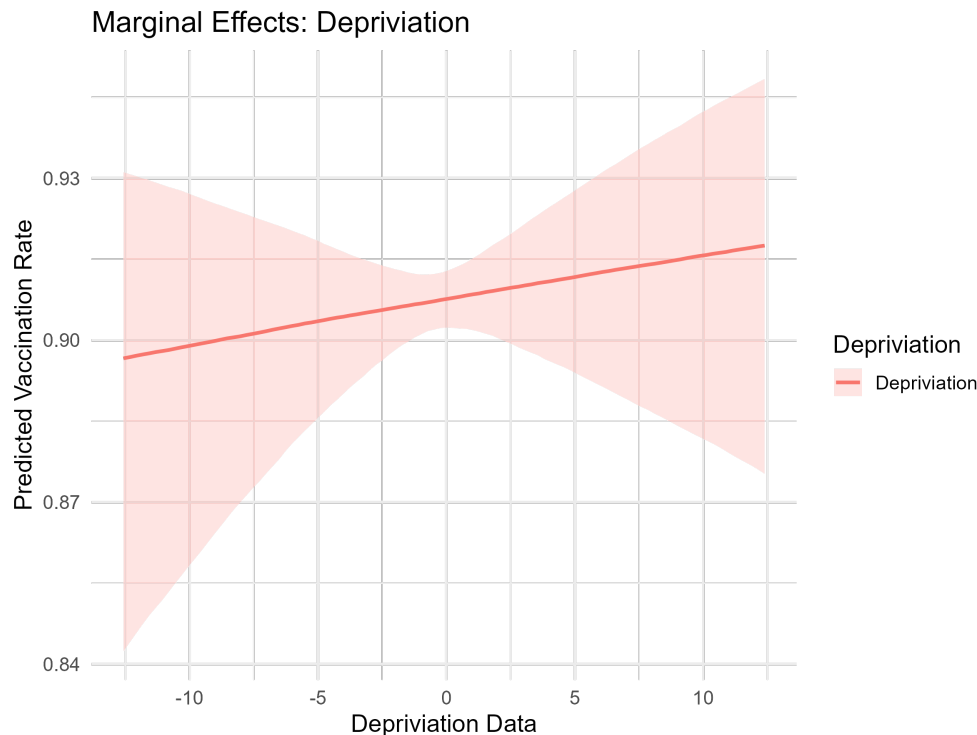


Figure 4: Marginal Effect of Deprivation

We identified substantial between-LEA heterogeneity in vaccination achievements, with the asymptote random effect standard deviation of 0.81 (95% CI: 0.73-0.91) indicating considerable variation in maximum vaccination rates between areas. The base asymptote showed more modest variation with a standard deviation of 0.25 (95% CI: 0.22-0.29), while the timing of peak uptake demonstrated relatively low variation between LEAs and the inflection point random effect standard deviation was only 0.26 (95% CI: 0.23-0.30), suggesting moderately synchronized temporal patterns across LEAs despite substantial differences in final vaccination levels.

The demographic predictors revealed significant positive associations between all covariates and final vaccination rate achievement. We found that more 65-70 years olds with respect to the 71+ year olds exhibited a strongest positive effect among age groups. It was also identified that LEAs with higher male proportions achieved substantially higher maximum vaccination coverage. The lower asymptote at -3.44 (95% CI: -3.49 to -3.39) on the logit scale, represents the minimal initial vaccination rates at the program's outset. Our demographic model maintained temporal stability, with the inflection point remaining consistent and the scale parameter showing minimal variation.

4 Discussions

The adjusted base model demonstrated superior fit, as evidenced by the increased precision parameter ($\phi = 77.74$ vs. 57.18), indicating reduced unexplained variance. The persistence of age effects across both models underscores the robustness of demographic patterns in vaccination behaviour, while the emergence of educational effects highlights the importance of socioeconomic factors. The negligible impact of initial vaccination center accessibility suggests that supply-side constraints were not primary drivers of vaccination disparities in this context, pointing instead to demand-side factors related to demographic and socioeconomic characteristics. The progressive increase in precision parameters across models ($57.18 \rightarrow 77.74 \rightarrow 87.88$) demonstrates continued improvement in model fit with the addition of socioeconomic variables and spatial structure and the spatial model represents the most comprehensive specification, accounting for both measured covariates and unmeasured spatially-correlated factors. The stability of key demographic and educational effects across all model specifications provides strong evidence for their robustness, while the significant spatial correlation highlights the importance of geographic context in understanding vaccination behaviour.

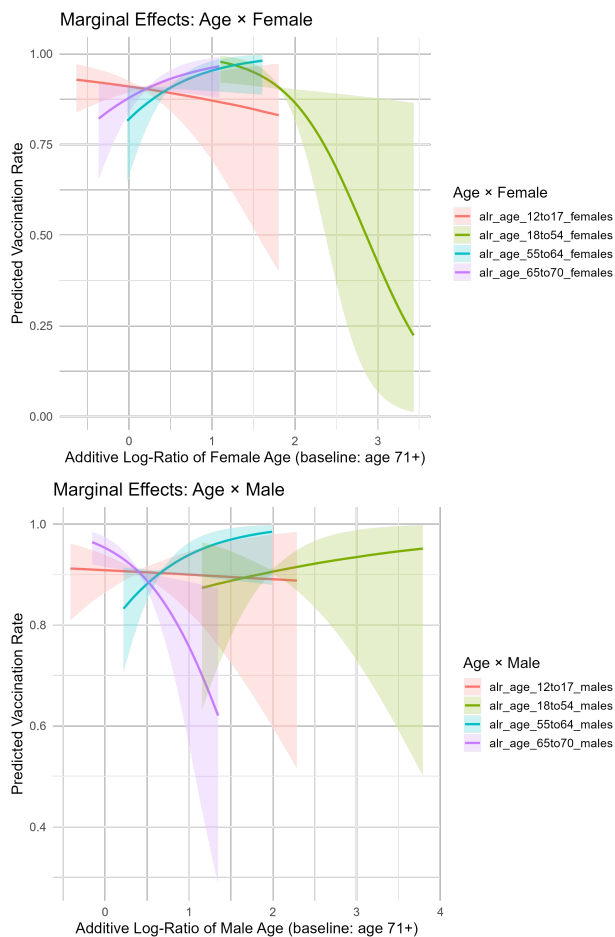


Figure 5: Marginal Effects of Age & Sex Interaction Model

patterns. The longitudinal nonlinear model revealed nuanced demographic dynamics in vaccination uptake that differ from patterns observed in cross-sectional analyses. Specifically, the Asym was significantly and positively associated with the relative proportions of males and all working-age and older adult subgroups (ages 18–54, 55–64, and 65–70), with the strongest effects observed for the older cohorts. These findings suggest that, although younger adults and males may have exhibited delayed uptake in earlier phases of the campaign—as suggested in prior models—they ultimately reached high levels of vaccination over time. The estimated population-level midpoint for uptake (x_{mid} approx 5.61 months) indicates broad temporal alignment across LEAs, reflecting a relatively coordinated national rollout. However, substantial variation in asymptotic values between LEAs ($sd(Asym) = 0.81$) underscores the importance of local demographic composition in shaping final uptake levels. That the scalar parameter governing growth rate ($scal = 1.36$) was fixed across areas reinforces the interpretation that differences in eventual uptake—rather than uptake speed—were more influenced by population structure than by supply-side constraints. Together, these results indicate that extended temporal observation is critical for capturing the full extent of vaccination behaviour: initial disparities in uptake timing do not necessarily translate into persistent inequalities in ultimate coverage. Based on the observed vaccination rate data, the increase-then-decline patterns of vaccination rate identified in LEAs such as South East Inner City, Maynooth, Galway City Central, Tramore-Waterford City West, and Kenmare suggest anomalies when compared to the typical sigmoidal vaccination uptake curve. A likely explanation for this pattern lies in changes to the underlying population denominators used in rate calculations. Since vaccination rates are computed relative to estimated local populations, any mid-series updates—such as those informed by the 2022 Census, student migration, or seasonal population shifts—could result in apparent declines despite stable or increasing absolute vaccination numbers. This effect is especially plausible in areas with transient populations, such as university towns (Maynooth, Galway), city centres with high mobility (South East Inner City), and tourism-influenced or rural areas (Kenmare, Skibbereen-West Cork), where even

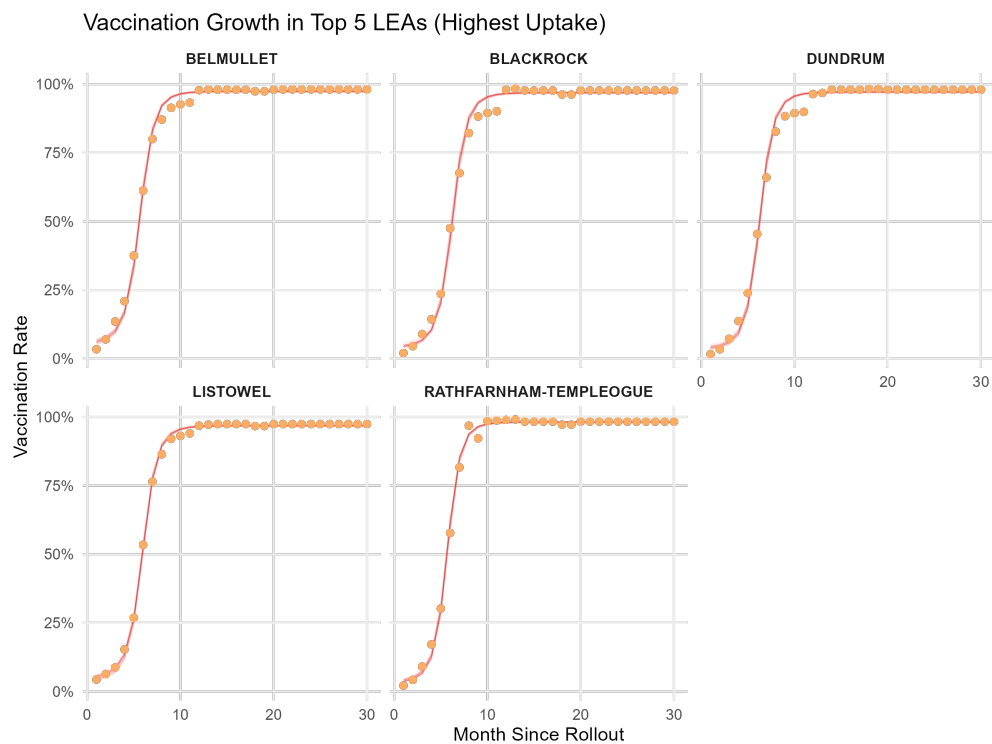


Figure 6: Vaccination Trend for Top 5 LEAs

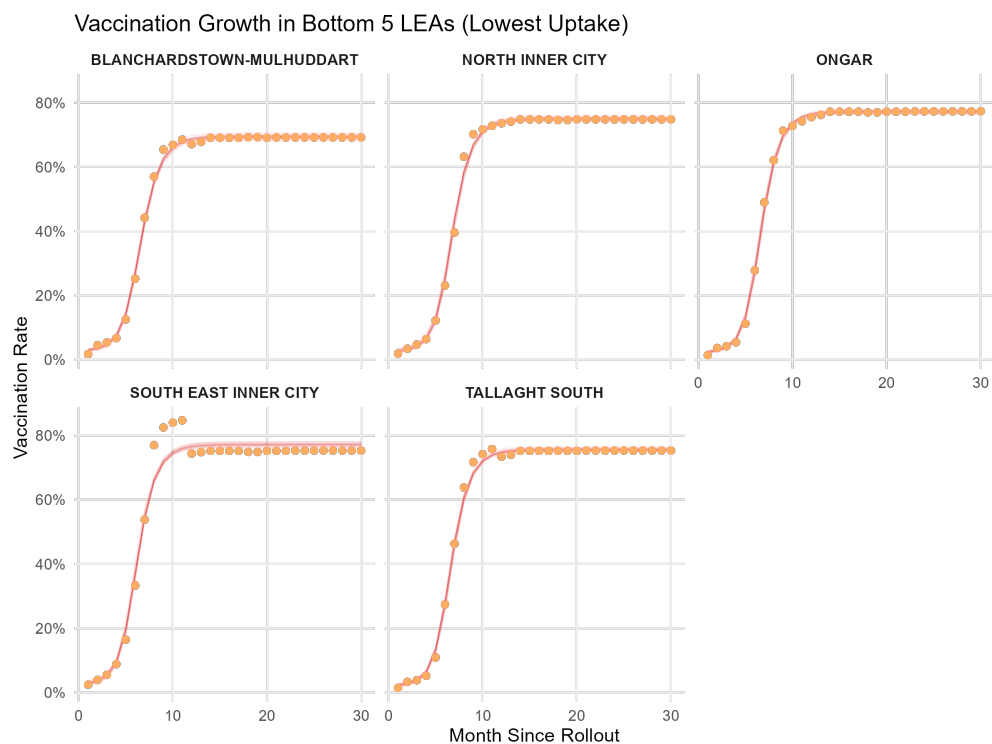


Figure 7: Vaccination Trend for Bottom 5 LEAs

small demographic changes can significantly distort percentage-based trends. As such, the observed dips are more likely to reflect population re-estimation or mobility rather than genuine decreases in vaccine uptake.

Our analysis relied on vaccination data reported only at the LEA level, not broken down by demographic subgroups or small area level data which was our biggest limitation. Population demographic characteristics were based on the 2022 Census and may not reflect the pandemic-period or vaccination roll out period scenario. Self-reported health status, used as a covariate, may be subject to bias. Our accessibility measure did not account for population demand or healthcare service’s capacity, limiting its interpretive power; while a 2SFCA approach could address these spatial limitations, it requires more granular data and computational resources beyond this study’s scope. Additionally, the analysis was restricted to the Republic of Ireland; cross-border vaccination—particularly in border regions—could have influenced local rates but remains unaccounted for in our models.

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